

Eco - Loop Building Agent

Autonomous Closed-Loop HVAC Optimization via EnergyPlus + Local LLM

Problem Statement Title:

Eco - Loop Building Agent

Theme:

Smart Automation / Sustainable Energy

PS Category:

Software

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What Eco - Loop Does:

- Links EnergyPlus (physics sim) with a local LLM (Gemma-4-E2B)
- Continuous loop: Sense → Reason → Act → Repeat, every hour
- Autonomously adjusts HVAC setpoints — zero human intervention

Innovation & Uniqueness:

- Fully local System – local LLM running on Nvidia rtx 2050 4gb vram
- Real physics co-simulation (not a rule engine)
- LLM output safety-wrapped by deterministic ASHRAE 55 guardrails
- 4× efficiency: 168 hourly decisions over 7 days in < 4 minutes

Traditional BMS vs EcoLoop AI

✗ Static schedules, no adaptation	✓ Dynamic setpoints, hourly AI decisions
✗ Ignores weather / occupancy	✓ Reads 6-hr weather forecast + live PMV
✗ Wastes energy on overcooling	✓ 23.34% cooling energy reduction
✗ No comfort feedback loop	✓ ASHRAE 55 PMV compliance: 98.2%

Technical Approach

Technologies Used:

- Simulation: EnergyPlus 26.1.0 + pyenergyplus PythonPlugin API
- Cognitive: Gemma-4-E2B — local LLM via llama.cpp (OpenAI REST)
- Language: Python 3.11 (src/ecoloop package)
- Data: EPW weather files, SQLite (eplusout.sql), JSONL log
- Dashboard: HTML + Chart.js (BMS Control Console)

Key Design Decisions:

- Hourly gating → 4× fewer LLM calls (672 → 168 over 7 days)
- max_tokens=128 → ~1.5s inference per call
- Fallback to last_good setpoints on timeout — never crashes EnergyPlus
- Skip actuation during design-day sizing (env_num < 3 guard)

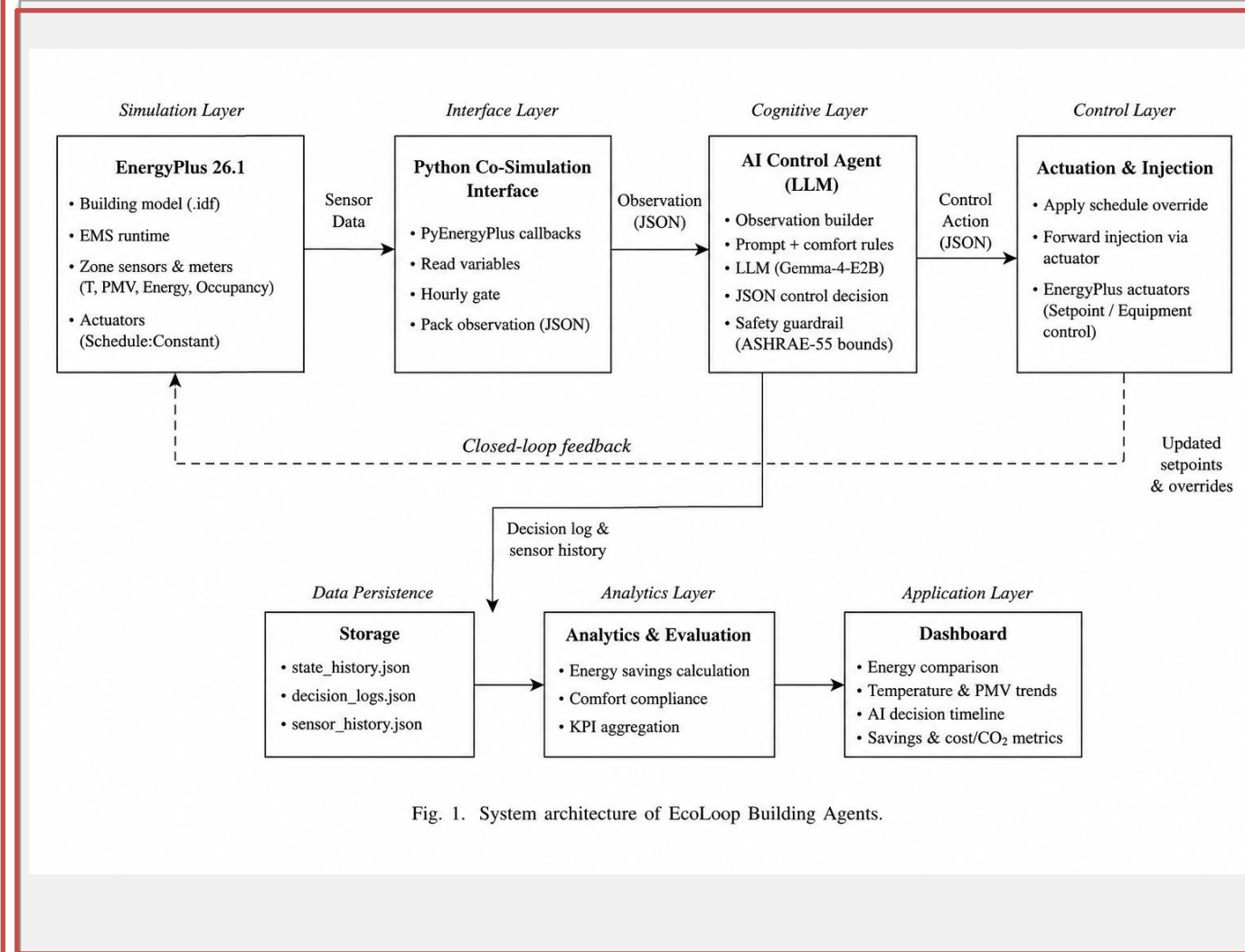


Fig. 1. System architecture of EcoLoop Building Agents.

Feasibility:

- Fully implemented & tested — 7-day Chicago summer simulation (July 1–7)
- Runs end-to-end in < 4 minutes on consumer hardware
- No internet or cloud required — 100% locally deployable without external api support

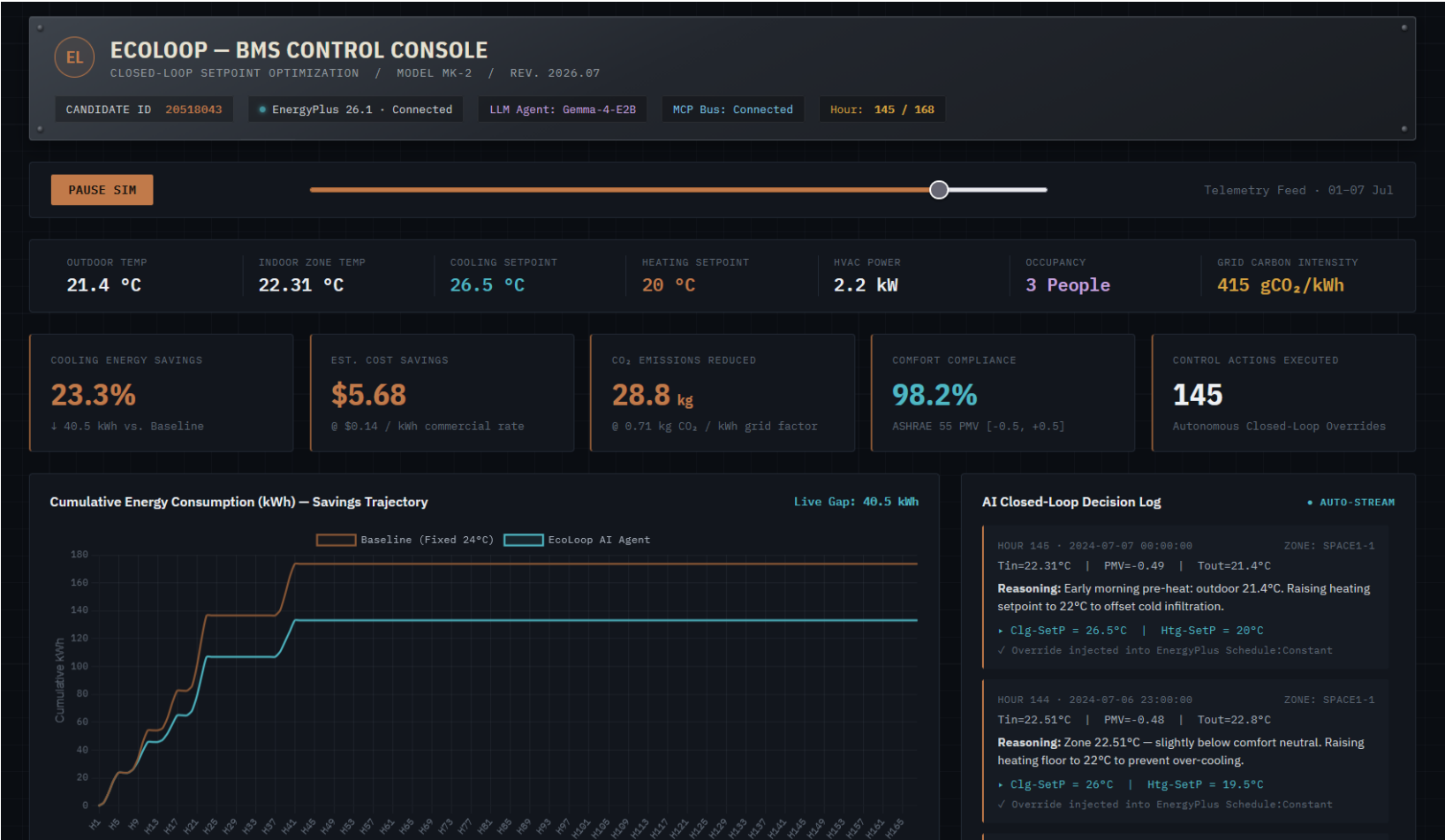
Challenge	Risk	Mitigation Strategy
LLM outputs malformed JSON	Simulation crash	Regex fallback + last_good setpoint reuse
LLM too slow for 672 timesteps	Unusable runtime (>50 min)	Hourly gating — 168 calls, < 4 min total
Setpoints outside safe HVAC range	Equipment damage / instability	Deterministic ASHRAE 55 clamp layer
LLM over-corrects (hunting)	Oscillating setpoints	3-step rolling decision memory in prompt
Design-day sizing interference	Corrupted HVAC sizing	env_num < 3 guard skips actuation

Artifacts — Results, Code & Dashboard



Quantitative Results (July 1–7, Chicago Summer)

Metric	Baseline	EcoLoop AI	Change
Cooling Energy	173.70 kWh	133.16 kWh	-23.34%
Avg PMV Index	-0.66	-0.55	+16% toward comfort
Avg Zone Temp	22.96 °C	23.35 °C	Less overcooling
AI Decisions Made	0	168	Hourly
Total Runtime	—	< 4 min	7 days simulated



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| 1. EnergyPlus Documentation | U.S. Department of Energy — energyplus.net/documentation |
| 2. ASHRAE Standard 55-2020 | Thermal Environmental Conditions for Human Occupancy — ashrae.org |
| 3. Fanger, P.O. (1970) | Thermal Comfort: Analysis and Applications in Environmental Engineering |
| 4. Gemma (Google DeepMind, 2024) | Open-weight LLM — ai.google.dev/gemma |
| 5. llama.cpp | Local LLM inference, OpenAI-compatible API — github.com/gggerganov/llama.cpp |
| 6. pyenergyplus | EnergyPlus Python API — github.com/NREL/EnergyPlus |
| 7. Model Context Protocol (MCP) | Anthropic open standard for agentic tool calling — modelcontextprotocol.io |
| 8. Project Repository | https://github.com/prakhxr0/agentic-bms |